

Exercise 1: Design of a Binary to BCD Conversion Circuit

Tuesday, 21 April 2026

Assignment

Design a circuit that takes as input a 4-bit binary number $B = B_3 B_2 B_1 B_0$ and converts it to a BCD (Binary Coded Decimal) representation with 2 digits $D_1 D_0$.

Example: The number $B=1101$ (13) will be converted to $D_1=0001$ (1) $D_0=0011$ (3).

Use the solution from Laboratory Exercise 3.5: Binary to BCD Converter as the basis for this assignment.

For the input B , use the switches SW3-SW0 (as in the Laboratory Exercise), but for the circuit output, use the LEDs on the board for D_1 (one LED is sufficient to connect the least significant bit of D_1 , as D_1 can be either 0 or 1) and the 7-segment display for D_0 . Refer to the Basys3 board manual for the operation of the 7-segment display:

Basys3 Board Manual (<https://digilent.com/reference/programmable-logic/basys-3/reference-manual>)

To drive the 7-segment LED, see the example slides 1-47, 1-48: BCD-to-7 segment decoder.

Note: To display the number on the 7-segment LED 0, you need to activate the anode signal (AN0), i.e., drive it permanently to 1 (active high), and appropriately drive the common cathode signals (CA ... CG). Note that the cathode signals are active low..

Examination

- The assignment will be examined on Thursday, April 30, 2026, at 13.00 (during the laboratory class).
- You can test your assignment on the board either on the day of the previous laboratory class or on the examination day.
- Collaboration between students is allowed, but the examination and grading are individual.
- YOU DO NOT need to submit any documentation or design files. You must have your design files (the Vivado project) with you during the examination.
- You will be examined on this specific exercise (questions will be asked about the VHDL model you implemented), as well as on the design tool and the board you used.

For questions, contact:

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